



ARTIFICIAL INTELLIGENCE HAND GESTURE RECOGNITION USING BAG-OF-VISUAL-WORDS

Prepared by:

Student Name (ID)

Date



Agenda

1. Introduction

2. Related Research and Key Technology

3. Project Contents and Algorithm

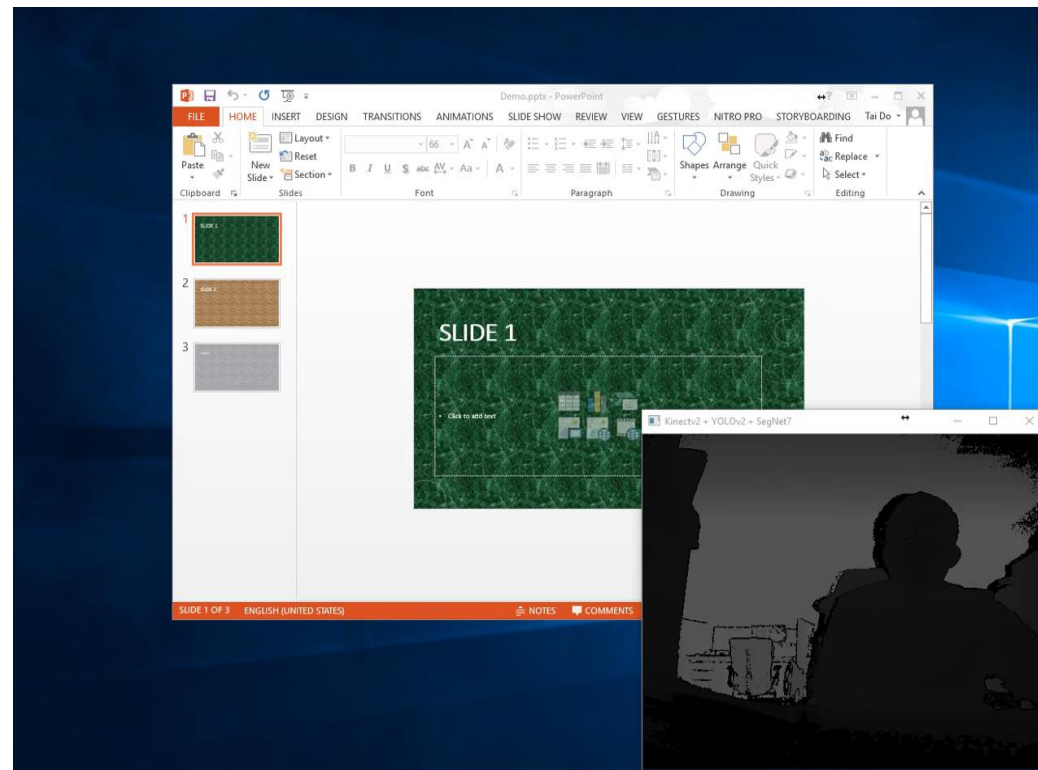
4. Experiment and Result

5. Conclusion



1. Introduction

- **Dynamic Hand Gesture Recognition:**
 - exploiting more the temporal features from the handshape sequence
 - challenging due to its small size, complexity, and self-occlusion
 - applying in human-machine interaction

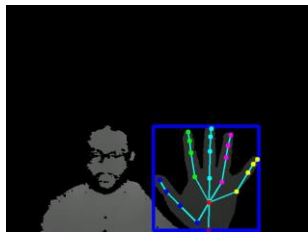




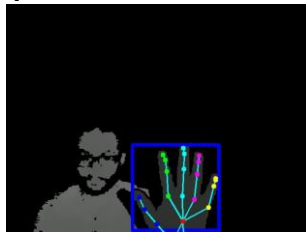
1. Introduction

- **Our problem:**

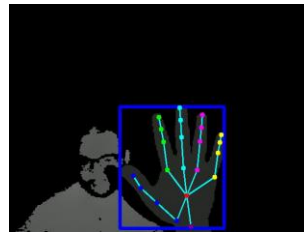
- Dynamic Hand Gesture Recognition on DHG Dataset
- Hand shape features based on Skeleton information and Depth image



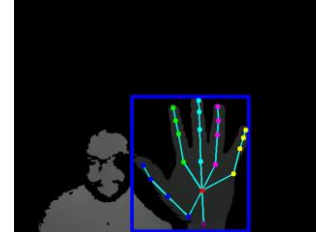
Grab



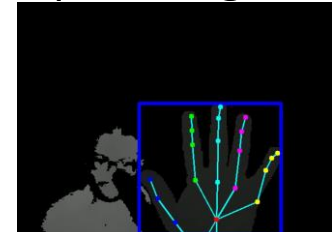
Tap



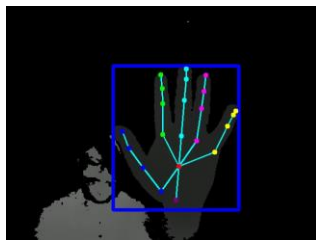
Expand



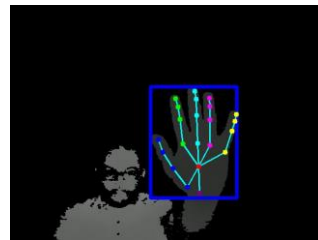
Pinch



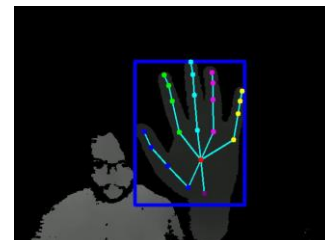
Rotation CW



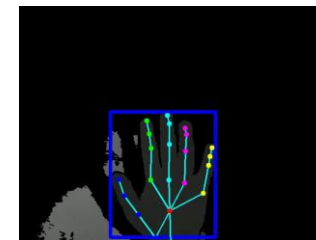
Rotation CCW



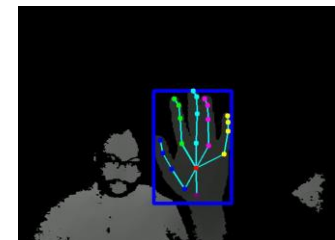
Swipe Right



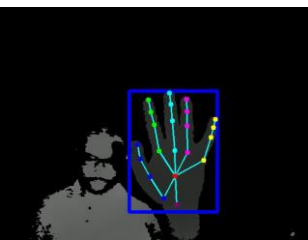
Swipe Left



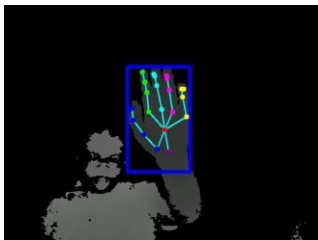
Swipe Up



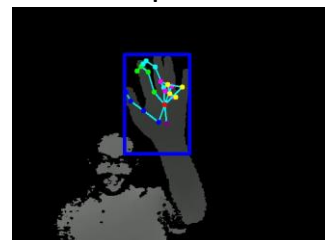
Swipe Down



Swipe X



Swipe V



Swipe +

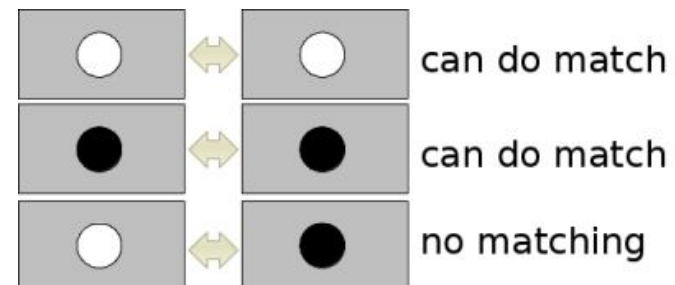
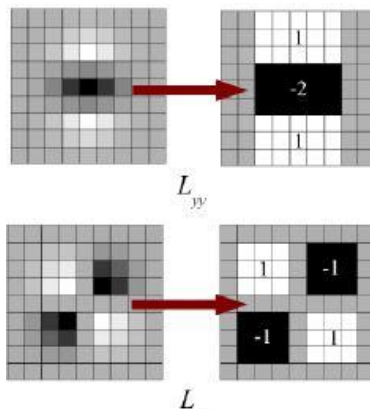
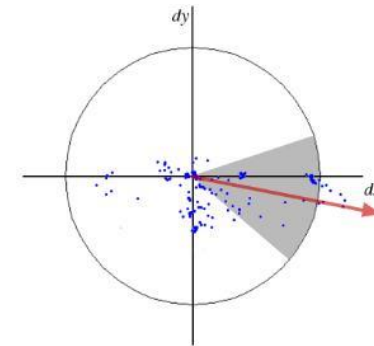
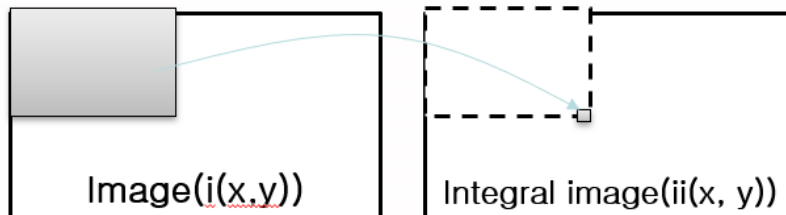


Shake



2. Related Research and Key Technology

- **Speeded Up Robust Features (SURF):**
 - approximates Laplacian of Gaussian with Box Filter using Integral image
 - feature description uses wavelet responses
 - sign of Laplacian for underlying interest point

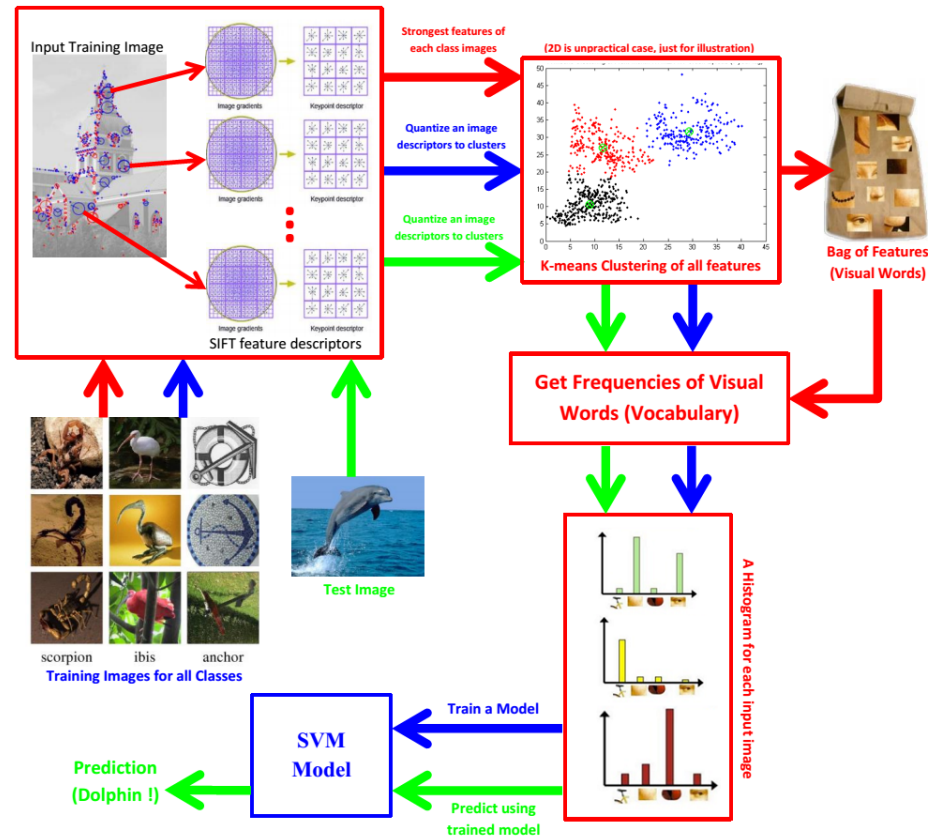




2. Related Research and Key Technology

- **Bag-Of-Visual Words:**

- An image is represented by a collection of “visual words” and their corresponding counts given a universal dictionary.

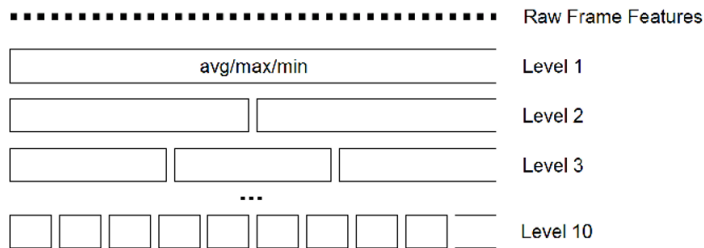




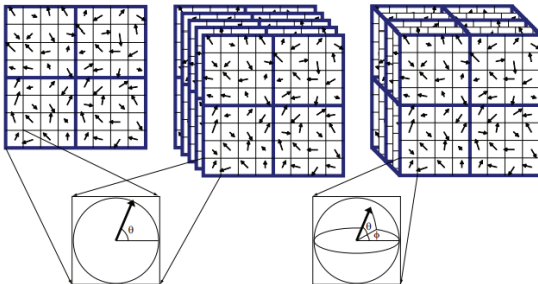
2. Related Research and Key Technology

• Temporal Model:

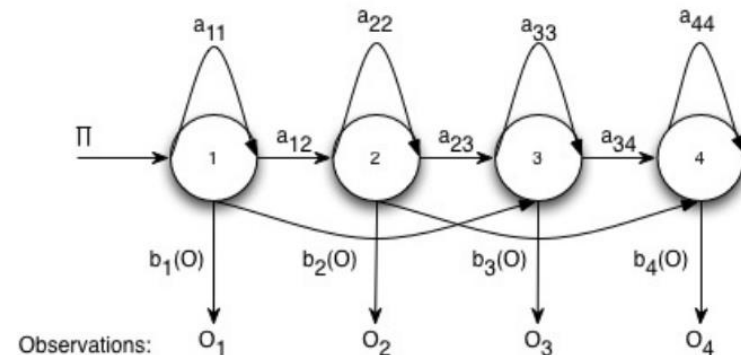
- descriptors which carry spatial and temporal information
- modeling sequences of spatial descriptors via temporal classifiers



Temporal Pyramid Model¹



3D SIFT²



a_{ij} - state transition probabilities

$b_j(O)$ - observation probabilities

Hidden Markov Model

[1] Choi, Jaesik, Won J. Jeon, and Sang-Chul Lee. "Spatio-temporal pyramid matching for sports videos." Proceedings of the 1st ACM international conference on Multimedia information retrieval. ACM, 2008

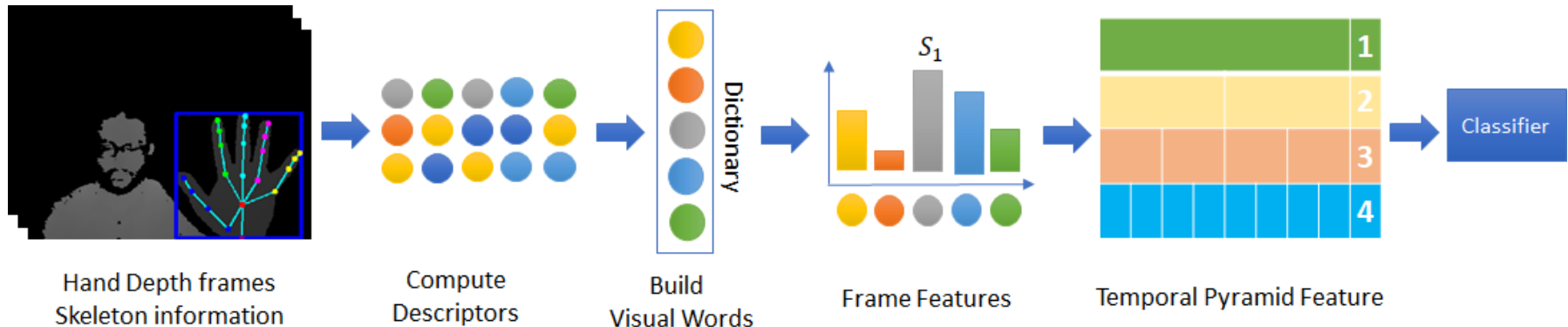
[2] Scovanner, Paul, Saad Ali, and Mubarak Shah. "**A 3-dimensional sift descriptor and its application to action recognition.**" Proceedings of the 15th ACM international conference on Multimedia. ACM, 2007.



3. Project Contents and Algorithm

- **Overview:**

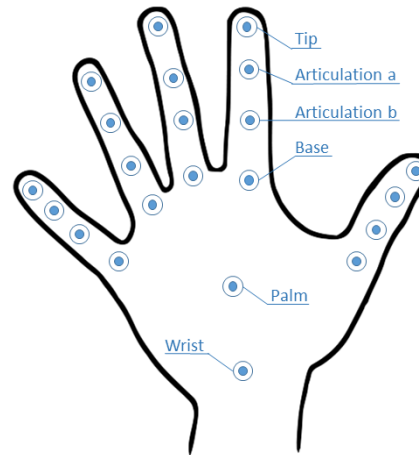
- (1) Extracting hand shape feature: skeleton information, depth image
- (2) Clustering local patches using k-means and Gaussian Mixture Model
- (3) Computing frame feature from visual-words
- (4) Extracting temporal pyramid feature
- (5) Classifying the gesture





3. Project Contents and Algorithm

- Hand shape feature from skeleton and depth:

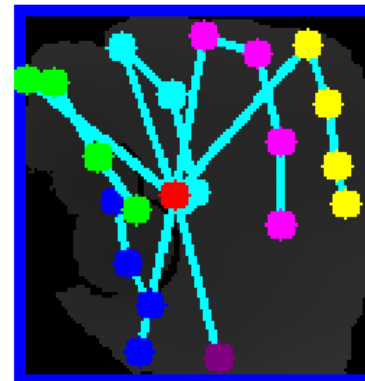


DHG Dataset annotation

Depth Image



Gesture





3. Project Contents and Algorithm

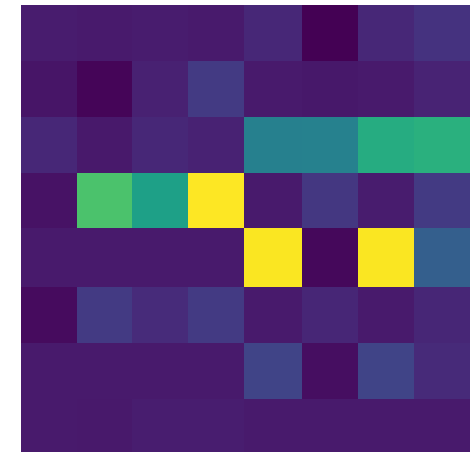
- **Depth image descriptor:**
 - Extracting keypoints and descriptors using SURF
 - 64 dimension feature vector



Keypoint



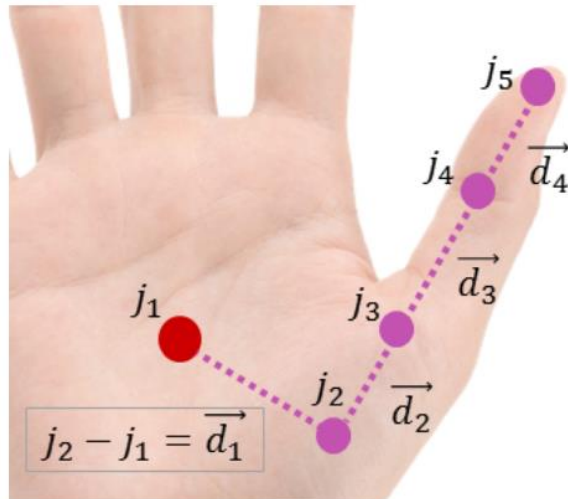
Descriptor



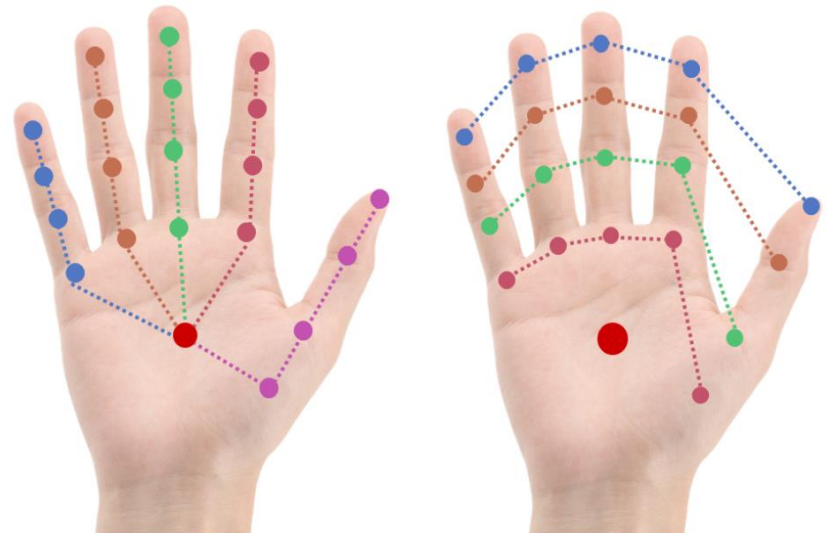


3. Project Contents and Algorithm

- **Skeleton information descriptor:**
 - Generate join pairs from 22 joints $\rightarrow$ 225 pairs
 - Calculate distance of a pair $\rightarrow$ 450-dimension feature vector



joint tuple distances

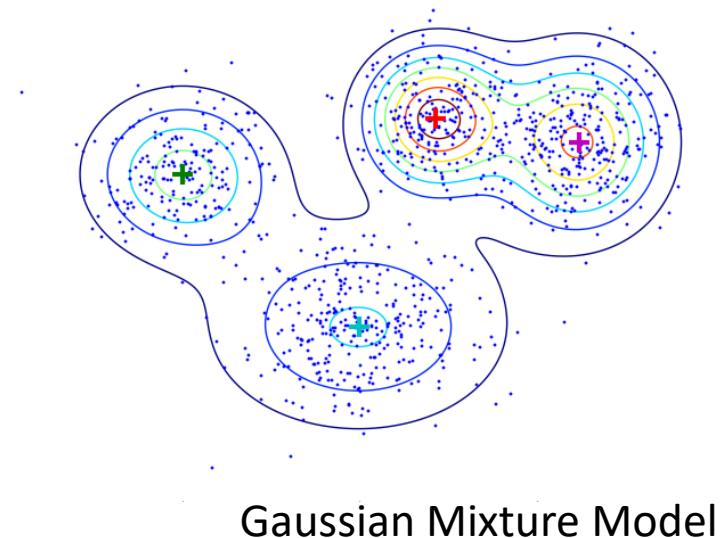
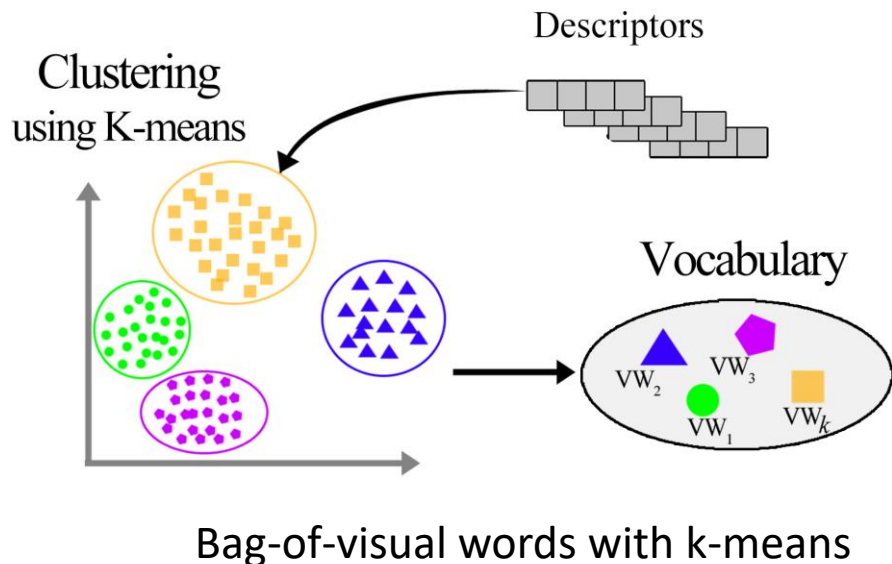


nine tuples sample for hand shape structure



3. Project Contents and Algorithm

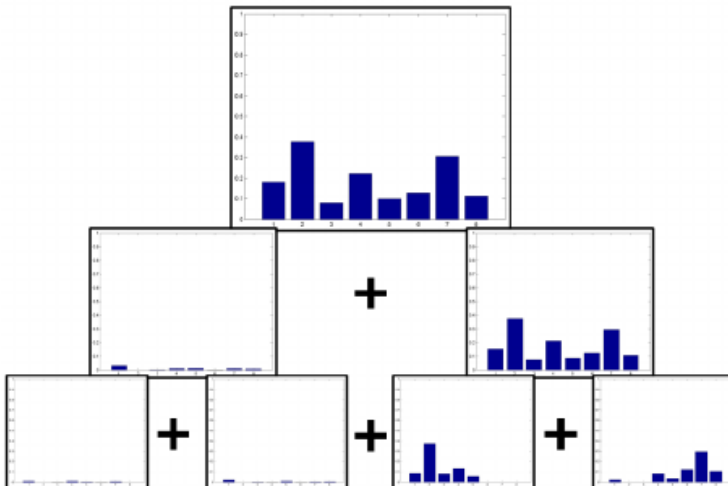
- **Clustering with Gaussian Mixture Model and k-means:**
 - Choose $k = 500$ visual words as number of clusters
 - It takes 4 hours for clustering
 - With GMM, only use sub-set of data for clustering



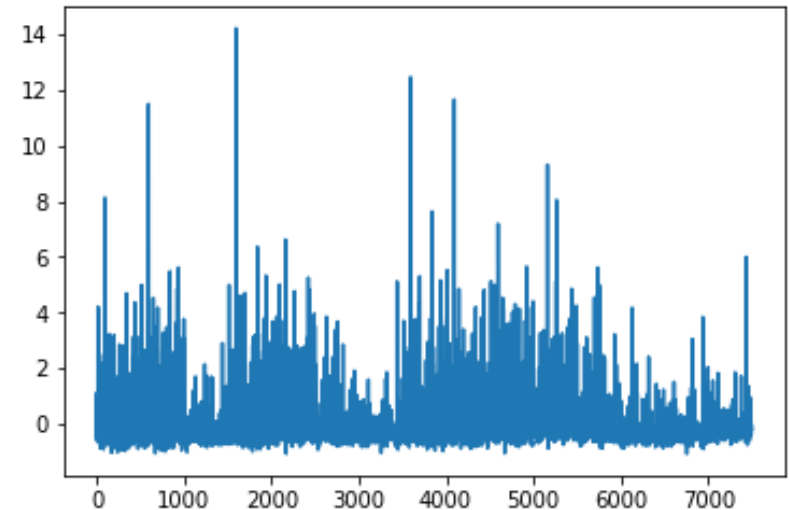


3. Project Contents and Algorithm

- **Temporal Pyramid Feature Extraction:**
 - Choose $n = 4$ level sub-sequences
 - With 500 visual-words, extracting 7500-dimension feature vector
- **SVM Classification:**
 - $C = 100.0$, $\gamma = \text{"auto"}$, $\text{probability} = \text{"True"}$, $\text{kernel} = \text{"rbf"}$



3 level temporal pyramid feature



An example of sequence feature



4. Experiment and Result

- Proposed methods:**

Method	Feature	Clustering	Operator
1	Skeleton	K-means	
2	Skeleton	GMM	
3	Depth	K-means	
4	Depth	GMM	
5	Fusion		Max
6	Fusion		Average
7	Fusion		Weighted



4. Experiment and Result

- Skeleton Features:**

Average accuracy 55.18%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	40.5	2.7	2.7	35.1	0.0	2.7	5.4	0.0	0.0	10.8	0.0	0.0	0.0	0.0
Tap	6.5	58.7	0.0	0.0	0.0	0.0	2.2	2.2	4.3	2.2	8.7	4.3	4.3	6.5
Expand	2.6	0.0	66.7	2.6	0.0	0.0	0.0	0.0	0.0	25.6	0.0	0.0	0.0	2.6
Pinch	34.0	2.1	2.1	44.7	0.0	0.0	0.0	0.0	2.1	0.0	14.9	0.0	0.0	0.0
Rotation CW	0.0	2.9	2.9	0.0	55.9	0.0	0.0	8.8	0.0	5.9	23.5	0.0	0.0	0.0
Rotation CCW	1.9	0.0	0.0	0.0	1.9	71.2	1.9	1.9	1.9	15.4	1.9	0.0	0.0	0.0
Swipe Right	2.3	4.7	2.3	4.7	7.0	0.0	62.8	0.0	0.0	9.3	0.0	4.7	2.3	0.0
Swipe Left	0.0	7.9	0.0	0.0	2.6	2.6	0.0	47.4	10.5	0.0	21.1	0.0	0.0	7.9
Swipe Up	2.3	2.3	2.3	0.0	2.3	0.0	4.5	0.0	34.1	0.0	40.9	4.5	2.3	4.5
Swipe Down	10.0	7.5	0.0	0.0	0.0	2.5	2.5	7.5	2.5	47.5	12.5	5.0	2.5	0.0
Swipe X	2.8	0.0	0.0	0.0	2.8	0.0	2.8	0.0	2.8	0.0	75.0	0.0	13.9	0.0
Swipe V	2.9	2.9	0.0	2.9	0.0	0.0	0.0	2.9	2.9	0.0	26.5	55.9	2.9	0.0
Swipe +	0.0	0.0	0.0	0.0	0.0	0.0	16.0	0.0	0.0	4.0	16.0	4.0	56.0	4.0
Shake	2.2	4.4	0.0	2.2	0.0	0.0	2.2	2.2	0.0	2.2	13.3	8.9	6.7	55.6

Gaussian Mixture Model

Average accuracy 53.39%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	40.5	0.0	5.4	40.5	0.0	0.0	0.0	0.0	2.7	2.7	5.4	2.7	0.0	0.0
Tap	4.3	60.9	2.2	4.3	0.0	0.0	0.0	0.0	13.0	2.2	13.0	0.0	0.0	0.0
Expand	2.6	5.1	66.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	23.1	0.0	2.6	0.0
Pinch	34.0	2.1	4.3	40.4	0.0	0.0	0.0	2.1	2.1	0.0	14.9	0.0	0.0	0.0
Rotation CW	0.0	0.0	2.9	0.0	64.7	0.0	0.0	5.9	0.0	0.0	23.5	0.0	0.0	2.9
Rotation CCW	3.8	0.0	1.9	0.0	0.0	76.9	0.0	1.9	1.9	0.0	13.5	0.0	0.0	0.0
Swipe Right	2.3	2.3	2.3	2.3	9.3	0.0	48.8	0.0	2.3	4.7	9.3	2.3	11.6	2.3
Swipe Left	0.0	2.6	0.0	0.0	7.9	2.6	0.0	50.0	7.9	5.3	18.4	2.6	0.0	2.6
Swipe Up	0.0	15.9	4.5	0.0	0.0	0.0	6.8	2.3	31.8	2.3	31.8	0.0	0.0	4.5
Swipe Down	10.0	12.5	0.0	2.5	0.0	2.5	0.0	7.5	5.0	50.0	7.5	2.5	0.0	0.0
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	5.6	0.0	2.8	2.8	72.2	0.0	13.9	2.8
Swipe V	0.0	0.0	2.9	8.8	2.9	0.0	0.0	0.0	5.9	0.0	26.5	47.1	0.0	5.9
Swipe +	0.0	0.0	0.0	0.0	0.0	0.0	8.0	0.0	0.0	4.0	28.0	8.0	52.0	0.0
Shake	2.2	8.9	6.7	0.0	2.2	0.0	0.0	8.9	4.4	0.0	17.8	4.4	0.0	44.4

K-means



4. Experiment and Result

- Depth Features:**

Average accuracy 71.96%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	45.9	8.1	13.5	24.3	0.0	0.0	0.0	0.0	0.0	8.1	0.0	0.0	0.0	0.0
Tap	6.5	78.3	6.5	0.0	6.5	0.0	0.0	0.0	0.0	2.2	0.0	0.0	0.0	0.0
Expand	0.0	0.0	92.3	2.6	0.0	0.0	0.0	0.0	5.1	0.0	0.0	0.0	0.0	0.0
Pinch	10.6	0.0	10.6	68.1	0.0	4.3	0.0	0.0	6.4	0.0	0.0	0.0	0.0	0.0
Rotation CW	2.9	0.0	11.8	0.0	73.5	2.9	2.9	0.0	2.9	2.9	0.0	0.0	0.0	0.0
Rotation CCW	1.9	1.9	17.3	0.0	7.7	63.5	0.0	1.9	0.0	3.8	1.9	0.0	0.0	0.0
Swipe Right	0.0	0.0	7.0	0.0	4.7	2.3	72.1	2.3	2.3	4.7	0.0	0.0	0.0	4.7
Swipe Left	0.0	0.0	7.9	0.0	2.6	2.6	10.5	60.5	5.3	7.9	0.0	0.0	0.0	2.6
Swipe Up	0.0	2.3	6.8	0.0	2.3	0.0	4.5	2.3	70.5	0.0	4.5	0.0	2.3	4.5
Swipe Down	5.0	5.0	0.0	2.5	2.5	0.0	0.0	2.5	2.5	75.0	2.5	2.5	0.0	0.0
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	83.3	5.6	11.1	0.0	0.0
Swipe V	0.0	0.0	11.8	2.9	0.0	0.0	2.9	0.0	2.9	2.9	0.0	76.5	0.0	0.0
Swipe +	0.0	0.0	4.0	4.0	0.0	0.0	0.0	0.0	4.0	12.0	0.0	72.0	4.0	0.0
Shake	0.0	0.0	4.4	0.0	6.7	0.0	0.0	6.7	0.0	2.2	2.2	0.0	0.0	77.8

Gaussian Mixture Model

Average accuracy 76.25%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	59.5	2.7	5.4	24.3	8.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tap	4.3	76.1	6.5	0.0	4.3	0.0	0.0	0.0	4.3	4.3	0.0	0.0	0.0	0.0
Expand	2.6	0.0	94.9	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pinch	14.9	0.0	8.5	72.3	2.1	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rotation CW	0.0	0.0	11.8	0.0	76.5	5.9	2.9	0.0	2.9	0.0	0.0	0.0	0.0	0.0
Rotation CCW	3.8	1.9	13.5	0.0	7.7	65.4	0.0	3.8	0.0	3.8	0.0	0.0	0.0	0.0
Swipe Right	0.0	4.7	4.7	0.0	4.7	0.0	72.1	0.0	4.7	2.3	0.0	2.3	0.0	4.7
Swipe Left	0.0	0.0	5.3	0.0	0.0	5.3	7.9	68.4	0.0	10.5	0.0	0.0	0.0	2.6
Swipe Up	0.0	0.0	4.5	4.5	2.3	0.0	4.5	6.8	68.2	0.0	4.5	2.3	2.3	0.0
Swipe Down	2.5	7.5	0.0	0.0	2.5	0.0	0.0	2.5	2.5	77.5	2.5	0.0	0.0	2.5
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	5.6	0.0	0.0	0.0	91.7	0.0	2.8	0.0
Swipe V	0.0	0.0	5.9	0.0	0.0	0.0	0.0	0.0	2.9	0.0	91.2	0.0	0.0	0.0
Swipe +	0.0	0.0	4.0	0.0	0.0	0.0	0.0	0.0	4.0	8.0	0.0	84.0	0.0	0.0
Shake	0.0	4.4	4.4	0.0	2.2	0.0	2.2	2.2	0.0	0.0	4.4	0.0	0.0	80.0

K-means



4. Experiment and Result

- Max and Average Fusion:

Average accuracy 80.89%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	62.2	2.7	0.0	29.7	0.0	2.7	0.0	0.0	0.0	0.0	2.7	0.0	0.0	0.0
Tap	4.3	82.6	0.0	0.0	2.2	0.0	0.0	0.0	2.2	0.0	4.3	0.0	2.2	2.2
Expand	0.0	2.6	89.7	2.6	0.0	0.0	0.0	0.0	2.6	0.0	0.0	0.0	2.6	0.0
Pinch	12.8	2.1	8.5	66.0	0.0	2.1	0.0	0.0	2.1	0.0	6.4	0.0	0.0	0.0
Rotation CW	0.0	0.0	2.9	0.0	76.5	2.9	2.9	5.9	2.9	2.9	2.9	0.0	0.0	0.0
Rotation CCW	1.9	0.0	1.9	0.0	1.9	90.4	0.0	0.0	1.9	0.0	1.9	0.0	0.0	0.0
Swipe Right	0.0	0.0	0.0	0.0	2.3	0.0	86.0	0.0	2.3	2.3	0.0	2.3	0.0	4.7
Swipe Left	0.0	0.0	2.6	0.0	2.6	0.0	10.5	76.3	0.0	2.6	2.6	0.0	0.0	2.6
Swipe Up	0.0	2.3	2.3	2.3	0.0	0.0	2.3	6.8	75.0	0.0	9.1	0.0	0.0	0.0
Swipe Down	0.0	7.5	0.0	2.5	2.5	2.5	0.0	5.0	0.0	77.5	0.0	0.0	0.0	2.5
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	2.8	0.0	0.0	0.0	86.1	5.6	5.6	0.0
Swipe V	0.0	0.0	0.0	0.0	2.9	0.0	0.0	0.0	0.0	2.9	2.9	91.2	0.0	0.0
Swipe +	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	8.0	0.0	88.0	0.0
Shake	0.0	2.2	4.4	0.0	4.4	0.0	0.0	0.0	0.0	0.0	2.2	0.0	0.0	86.7

Max Fusion

Average accuracy 82.50%

True label \ Predicted label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	67.6	2.7	5.4	21.6	0.0	0.0	0.0	0.0	0.0	0.0	2.7	0.0	0.0	0.0
Tap	2.2	87.0	4.3	0.0	2.2	0.0	0.0	0.0	4.3	0.0	0.0	0.0	0.0	0.0
Expand	0.0	2.6	94.9	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pinch	14.9	0.0	6.4	70.2	0.0	2.1	0.0	0.0	0.0	0.0	6.4	0.0	0.0	0.0
Rotation CW	0.0	0.0	8.8	0.0	79.4	0.0	2.9	2.9	2.9	2.9	0.0	0.0	0.0	0.0
Rotation CCW	1.9	0.0	1.9	0.0	1.9	92.3	0.0	0.0	1.9	0.0	0.0	0.0	0.0	0.0
Swipe Right	0.0	0.0	0.0	0.0	0.0	0.0	86.0	0.0	2.3	4.7	0.0	2.3	0.0	4.7
Swipe Left	0.0	0.0	0.0	0.0	2.6	2.6	7.9	76.3	0.0	5.3	2.6	0.0	0.0	2.6
Swipe Up	0.0	2.3	2.3	2.3	0.0	0.0	2.3	4.5	77.3	0.0	6.8	2.3	0.0	0.0
Swipe Down	0.0	10.0	0.0	2.5	0.0	0.0	0.0	2.5	2.5	77.5	2.5	0.0	0.0	2.5
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	91.7	2.8	5.6	0.0
Swipe V	0.0	0.0	2.9	0.0	2.9	0.0	0.0	0.0	0.0	2.9	2.9	88.2	0.0	0.0
Swipe +	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	0.0	4.0	12.0	0.0	80.0	0.0
Shake	0.0	2.2	4.4	0.0	4.4	0.0	0.0	2.2	0.0	0.0	2.2	0.0	0.0	84.4

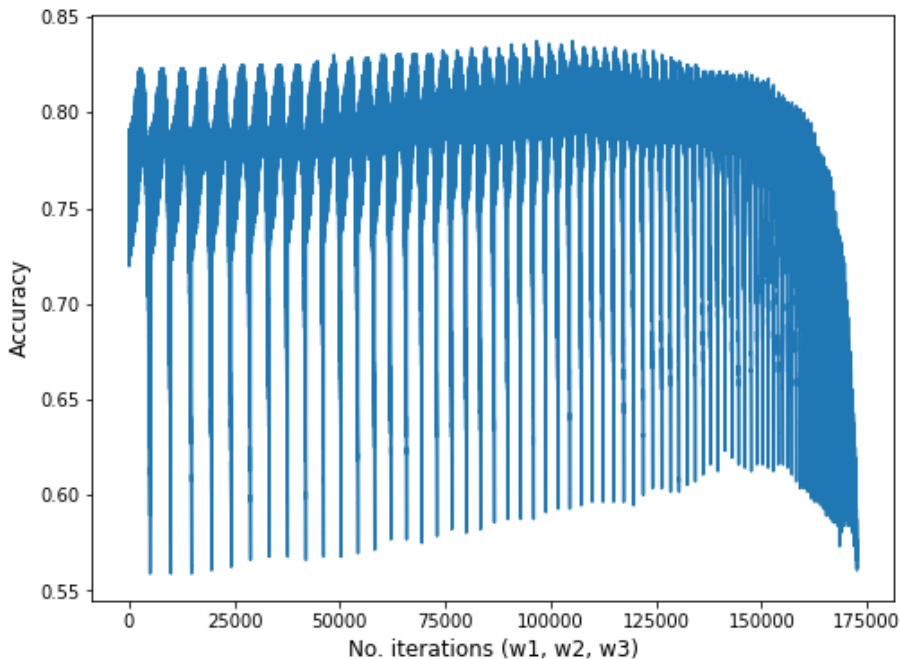
Average Fusion



4. Experiment and Result

- Weighted Fusion:**

$$w_1 \cdot \text{model}_1 + w_2 \cdot \text{model}_2 + w_3 \cdot \text{model}_3 + (1 - w_1 - w_2 - w_3) \cdot \text{model}_4$$



Search Params

Average accuracy 83.75%

True label	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
Grab	73.0	2.7	2.7	18.9	0.0	0.0	0.0	0.0	0.0	2.7	0.0	0.0	0.0	0.0
Tap	2.2	89.1	4.3	0.0	2.2	0.0	0.0	0.0	2.2	0.0	0.0	0.0	0.0	0.0
Expand	0.0	0.0	97.4	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pinch	12.8	0.0	6.4	72.3	0.0	2.1	0.0	0.0	0.0	4.3	2.1	0.0	0.0	0.0
Rotation CW	0.0	0.0	8.8	0.0	82.4	0.0	2.9	0.0	2.9	2.9	0.0	0.0	0.0	0.0
Rotation CCW	1.9	0.0	1.9	0.0	1.9	88.5	0.0	3.8	1.9	0.0	0.0	0.0	0.0	0.0
Swipe Right	0.0	0.0	2.3	0.0	0.0	0.0	86.0	0.0	2.3	4.7	0.0	0.0	0.0	4.7
Swipe Left	0.0	0.0	0.0	0.0	0.0	0.0	7.9	73.7	0.0	13.2	2.6	0.0	0.0	2.6
Swipe Up	0.0	0.0	4.5	2.3	2.3	0.0	2.3	4.5	77.3	0.0	2.3	2.3	0.0	2.3
Swipe Down	0.0	7.5	0.0	2.5	2.5	0.0	0.0	2.5	0.0	82.5	0.0	0.0	0.0	2.5
Swipe X	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	91.7	2.8	5.6	0.0
Swipe V	0.0	0.0	2.9	0.0	0.0	0.0	0.0	0.0	2.9	2.9	0.0	91.2	0.0	0.0
Swipe +	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	12.0	0.0	0.0	84.0	0.0
Shake	0.0	2.2	4.4	0.0	4.4	0.0	0.0	2.2	0.0	2.2	0.0	0.0	0.0	84.4
	Grab	Tap	Expand	Pinch	Rotation CW	Rotation CCW	Swipe Right	Swipe Left	Swipe Up	Swipe Down	Swipe X	Swipe V	Swipe +	Shake
	Predicted label													

Weighted Fusion



4. Experiment and Result

- Summary:**

Method	Feature	Clustering	Operator	Accuracy
1	Skeleton	K-means		53.39%
2	Skeleton	GMM		55.18%
3	Depth	K-means		76.25%
4	Depth	GMM		71.96%
5	Fusion		Max	80.89%
6	Fusion		Average	82.50%
7	Fusion		Weighted	83.75%



5. Conclusion

- Propose dynamic hand gesture recognition
- Base on hand shape feature with skeleton and depth information
- Use bag-of-visual-words with k-means and gaussian mixture model
- Extract temporal pyramid feature for sequence
- Recognition gesture with SVM Classification
- Ensemble results with maximum, average and weighted operator
- Achieve good accuracy on DHG Dataset



THANKS FOR LISTENING!